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Belowground morphology and population dynamics of two forest understory herbs of contrasting growth form

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Abstract: Forest understory herbs exhibit a large range of variation in morphology and life history. Here we expand the reported range of variation by describing the belowground structures of two very different species, *Xerophyllum tenax* and *Chimaphila umbellata*. We excavated individuals in forests of the Cascade Mountains, Pacific Northwest, USA. *Xerophyllum tenax* has short rhizomes, but an extensive root system that is exceptionally large among forest understory species. The roots reach 4 m in length and may occupy an area 50 times that of the aboveground canopy. In contrast, *Chimaphila umbellata* has very small roots, but an extensive rhizome system. The largest plant we excavated had 57 m of connected rhizomes and still had a seedling source. Both species have long-lived individuals but differ in response to disturbance. Based on monitoring of 151 permanent 1 m^2 plots in an old-growth forest, *X. tenax* increased only minimally in density over 40 years following tephra deposition from the 1980 eruption of Mount St. Helens, whereas density of *C. umbellata* increased substantially. The very different morphology of these two species highlights the large range of variation found among forest herbs, which needs to be considered when examining the forest understory.

Key words: Chimaphila umbellata, forest herbs, forest understory plants, Mount St. Helens, rhizomes, root systems, volcanic tephra, *Xerophyllum tenax*

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Introduction

Understory herbs are an important component of many forest ecosystems and often provide the majority of the vascular species in temperate forests (Bierzychudek 1982; Whigham 2004; Gilliam 2007). These species constitute a major functional group, which share a similar habitat, occurring in the forest understory where light levels are typically low and where competition from trees for soil-based resources is also severe (McCune 1986; Coomes and Grubb 2000; Lindh et al. 2003). Understory herb species do differ, though, in a variety of respects, including ability to grow in low light and response to disturbance (Roberts 2004; Zobel and Antos 2009), with a gradient from species that occur primarily in deep shade to those that grow best after disturbance but can persist under a canopy. Species also differ greatly in the amount**s** of flowering and seedling production (Antos and Zobel 1986; Lindh 2005; Zobel and Antos 2007, 2016), and some forest herbs flower primarily beneath canopy gaps while others persist and reproduce in shade (Valverde and Silvertown 1998; Lindh 2005). Both species we studied here have limited flowering in the forest understory: *Chimaphila umbellata* flowered less (5% of shoots flowering) than most other major herb species and flowering was not observed for *Xerophyllum tenax* in our plots (Zobel and Antos 2016). In addition, forest understory herbs differ in their source of energy, with some species being partially dependent on fungi (Jacquemyn and Merckx 2019), including *C. umbellata*. Thus**,** forest herbs differ in many characteristics related to physiology and reproduction.

In addition, morphological variation among forest herbs, especially in belowground structures, is pronounced and perhaps their most extreme form of variation. Some species spread rapidly via aboveground stolons, which may be especially important following disturbance (Zobel and Antos 1997, 2009, 2017). However, clonal spread belowground via rhizomes is more

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typical of forest herbs, and species with perennating organs belowground range from those with extensive rhizome systems to species lacking vegetative spread (Antos and Zobel 1984, 1985*b*; Lezberg et al. 1999; Six and Halpern 2008; Moola and Vasseur 2009). In addition**,** root systems differ substantially among forest herbs, although most species have small roots (Antos and Zobel 1984, 1985*b*; Lezberg et al. 1999). For example, the maximum root length found among nine species in similar forests was less than 0.45 m (Antos and Zobel 1984). Moreover, species with very similar aboveground morphology can have very different structures belowground (Antos 1988). Among forest herbs with extensive clonal growth, species differ in the temporal persistence of connections among ramets, from only one year (Piqueras and Klimeš 1998) to decades (Antos and Zobel 1984, 1985*b*). Generally rhizomes die at the old ends and, when determining their age is possible, are less than 20 years old (Antos and Zobel 1984). Meanwhile, vegetation of the forest understory varies greatly at short distances (Burton et al. 2011; Chudomelová and Zelený 2020) as do light and soil resources, partly related to presence and species of canopy trees (Barbier et al. 2008; Chamagne et al. 2016). Plants with large clonal fragments can potentially integrate resources across microsites via persistent connections among ramets (Slade and Hutchings 1987; van Groenendael et al. 1996). In contrast, forest herbs that lack vegetative spread typically have small root systems (e.g., *Erythronium montanum*, Antos 1988) and can only use resources near the single shoot; they must rely on sexual reproduction to colonize areas opened by disturbance, and some such species produce abundant seedlings (Antos and Zobel 1986). Additionally, clonal growth may be pivotal to redevelopment of the forest understory after pronounced disturbance, so long as some plants survive. Thus population changes following disturbance can relate strongly to morphological differences among species. Clonal growth can also be very effective for local expansion, and many forest herbs have

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extensive clonal development, seldom flower, and appear to establish new genets rarely. In general, though, forest herbs have slow spread, both locally (Ehrlén and Eriksson 2000; Verheyen et al. 2003; Verheyen and Hermy 2004; Matlack 2005) and over long distances following environmental changes (Cain et al. 1998).

Here we describe the structure and long-term dynamics of two common herbs of the forest understory that extend the range of variation previously reported for forest herbs in the Pacific Northwest (Antos and Zobel 1984, 1985*b*; Lezberg et al. 1999, 2001; Nelson et al. 2007). This work was part of our studies to determine the mechanisms by which forest understory plants respond to tephra deposition from the 1980 eruption of Mount St. Helens (long-term plant responses are summarized in Zobel and Antos 2017, 2018; Fischer et al. 2016, 2019). Our specific objectives are to describe: (1) the growth form and persistence of *Xerophyllum tenax*, a large, grass-like monocot herb with limited clonal growth, but an extremely large root system; (2) the rhizome system and persistence of *Chimaphyla umbellata*, a partially mycoheterotrophic herb with exceptionally long-lived rhizomes; and (3) the contrasting patterns of change in importance of *X. tenax* and *C. umbellata* during 36-40 years of recovery from disturbance.

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Materials and methods

Species and study sites

Xerophyllum tenax (Melanthiaceae, beargrass) is a large herbaceous monocot with numerous stiff, narrow, evergreen leaves (Fig. 1a). The species can be abundant in the understory of some coniferous forests in the Pacific Northwest (Franklin and Dyrness 1973), as well as in open habitats from the lowlands along the coast to the subalpine (Hitchcock and Cronquist 2018). The leaves are used for basket making by indigenous peoples (Shebitz et al.

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2009; Peter et al. 2017). *Chimaphyla umbellata* (Ericaceae, prince's-pine or common pipsissewa) is an evergreen herb that is partially mycoheterothrophic as is typical of the subfamily Pyroleae (Matsuda et al. 2012; Johansson and Eriksson 2013; Jacquemyn and Merckx 2019) (Fig. 1b). Rhizomes of *C. umbellata* have average caloric content, relatively low staining for starch, and a relatively thick outer rind compared to several regional understory herbs (Zobel and Antos 1987).

To examine plant morphology and changes in morphology related to tephra burial, we excavated individuals of both species at a forest site ca. 23 km northeast of Mount St. Helens that received ca. 10 cm of tephra (aerially transported volcanic ejecta) in 1980. We also excavated plants of *C. umbellata* at a site in the central Western Cascades of Oregon with no recent tephra. This site was on the H.J. Andrews Experimental Forest and was near where we sampled plants of other species for a previous study (Antos and Zobel 1984). To examine demography and temporal changes, we used density and cover data from a site 58 km northeast of Mount St. Helens where we monitored long-term changes in the forest understory following the tephra disturbance (site SP, Fischer et al. 2019); this site received 4.5 cm of tephra in 1980. Site SP is in an old-growth forest dominated by large trees of several species with *Abies amabilis* and *Tsuga heterophylla* most important; *X. tenax* is the most abundant understory herb (estimated pre-eruption cover of 23.7 %, Zobel and Antos 1997).

Nomenclature follows Hitchcock and Cronquist (2018).

Morphological sampling

For *X. tenax* we excavated parts in the tephra for 11 plants during August and September, 1981, to determine if they had moved perennating structures upward into the tephra, as was the

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case for most species (Antos and Zobel 1985*a*, *b*). We excavated rhizomes in the pre-eruption soil and recorded rhizome depth for 10 of the 11 plants. For one of these plants we partially excavated the root system in the pre-eruption soil; the roots proved very numerous, long, difficult to trace, and slow to extract. On this plant, we counted all roots emanating from the short rhizomes and excavated ten of these roots, recording their length, diameter, and other characteristics. We also recorded the number and length of branch roots on four main root segments, each 1 m long.

For *C. umbellata* we completely excavated seven plants. Two of these formed a patch at the 10 cm tephra site and the other five formed small, distinct patches at the tephra-free site in Oregon. The complete rhizome systems of these seven plants were excavated, mapped, and the lengths of all rhizome segments recorded. All green shoots on these plants were described. Whorls of leaves, including scars from past leaves, appear to indicate annual increments on the aerial shoots so we counted whorls of leaves and leaf scars as an index of shoot age. We noted the condition of all rhizome ends. On one large plant at the tephra site we also recorded rhizome branch angles and number and length of roots on eight rhizome segments, each 0.5 m long. In general**,** the excavation procedures were similar to those we used previously (Antos and Zobel 1984, 1985*b*). We determined dry mass of plant parts (rhizomes plus roots, aerial stems, and leaves), and the mass of plant parts in tephra. We separated roots and rhizomes on four of the seven plants. Plant parts were washed, separated, and oven dried to constant weight at 70 C.

Our morphological sampling was limited because of the large size of the plants and difficulty of excavation. Among the many forest herb species we have excavated, these two species were exceptional, but extremely slow to excavate. Thus we did not attempt to obtain a large sample, but wanted to indicate their unusual size and belowground structures, which is

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relevant to appreciating the extent and nature of variation among forest herbs. However, because we chose a single typical shoot to begin each excavation, the values we report are likely to be common for these species.

Demographic sampling and analysis

At site SP we used permanently marked 1 m^2 plots, 101 with natural tephra and 50 experimentally cleared of tephra in 1980. We sampled the plots at this site periodically from 1980 to 2020, including density and cover of *X. tenax* and density (1980-2016) and cover (1980-2020) of aerial shoots *C. umbellata* (for specifics of sampling see Antos and Zobel 1985c; Zobel and Antos 1997, 2017; Fischer et al. 2019). To evaluate the significance of temporal trends in cover and density of each species following the 1980 tephra disturbance, we used paired t-tests for both natural tephra plots and plots cleared of tephra in 1980. We compared 1981 (the first year with post-clearing data) with 1983 to evaluate changes occurring soon after the 1980 tephra disturbance. To evaluate subsequent long-term changes, we compared 1983 with the last sampling date (2020, except 2016 for *C. umbellata* density). To examine possible differences between natural and cleared plots, we used t-tests for unequal variances for comparisons of cover and density in 1981, 1983, and 2020 (except 2016 for *C. umbellata* density). Significance in all tests was determined based on an α of 0.05, and a Bonferroni correction for number of tests where appropriate.

Results

Xerophyllum tenax

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The plants have short, branching rhizomes and a genet can have many ramets. We examined plants (genets) of a large size range, 1 to 35 ramets (or crowns, i.e., rhizome tips with tufts of leaves) per plant (Table 1). The rhizomes appear to elongate very slowly. Thus, even with many ramets, plant leaf canopies have limited spread; the 35-ramet plant we examined had a canopy diameter of only 90 cm (Table 1). The rhizomes are long-lived, and the ramets were still all connected on the plants we examined. New ramets are formed at the end of rhizomes that branch from older rhizomes near old ramets. Some of the branching occurs after a ramet develops a flowering stalk, but some branching occurs from ramets that do not appear to have ever flowered. Overall clonal spread is very limited in *X. tenax*.

Xerophyllum tenax has a dimorphic root system, with large, cord-like roots that have numerous short branch roots. Large roots extended much farther than the limited rhizome spread and leaf canopy. The plant for which we excavated roots (Plant 1, Table 1) was typical in size for its population, but had 184 large roots emanating from the rhizomes at the base of the 12 crowns. The roots reached a depth of at least 80 cm below the pre-eruption soil surface and may have gone much deeper. We excavated 10 of the 184 large roots and traced six of these roots to a natural end; two had dead tips and four had live tips. The lengths of the four roots with live tips were 85, 118, 147, and 405 cm. The large roots were generally straight, 1-3 mm in diameter, and rarely had major branch roots. However, small branch roots were common along the main roots; the branch roots were much thinner than the main roots and generally <10 cm long. We examined four root segments, each 1 m long, on three of the excavated large roots and found 28-42 small branch roots per meter with an average length of 2.6-5.2 cm among the four segments. The longest branch root was 21 cm. Given our small sample of the many large roots on this plant, we suspect that we did not excavate the longest root, but if we consider the longest

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excavated root as indicative of the rooting radius, this plant occupied (and could potentially extract soil resources from) about 50 m^2 belowground, which is greater than 50 times the area occupied by the plant aboveground (canopy diameter of only 85 cm, Table 1).

Over 40 years**,** we measured only limited change in ramet density (Fig. 2a) or species frequency (% of 1 m^2 plots occupied; Fig. 2b). Density did increase significantly from 1981 to 1983, but not afterwards in both natural and cleared plots (Table 2). Cover was reduced some during the first few years after the tephra deposit (Fig. 2c) due to burial of leaves; cover was significantly lower in natural than in cleared plots in 1981, but not in 1983 or 2020 (Table 2). Density and frequency of plants were similar in plots with and without tephra removed (Fig. 2a, b), indicating that 4.5 cm of tephra did not kill *X. tenax* although it reduced cover for a while (Fig. 2c). However, we found no evidence that the plants grew rhizomes or moved crowns upward into the tephra, in contrast to most species (Antos and Zobel 1985*a*, *b*). We did observe seedlings, but only in one plot at one sampling, and these did not persist. The minor early changes in density and frequency (Fig. 2) thusly relate to changes in ramet number or position, but not mortality or recruitment of genets. Overall the plants appear to be long lived with little mortality or recruitment. Photos taken at 1-m above plots in 1980 and 2016 similarly demonstrate stability among individuals and percent cover despite tephra-impacts, treefall disturbances, and variable light conditions (Fig. 3)

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Chimaphila umbellata

All of the plants we excavated had a distinct seedling source visible in their structure at the oldest end of the rhizome, where there was a large vertical root from which radiated three to five rhizomes (Table 3). This structure was distinct from any other place on the plants. All other

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rhizome junctions were simple branches with a source rhizome dividing into two rhizomes; a root often occurred at these junctions but these roots were much smaller than the root at the source and generally not vertical. On the largest plant (Tephra 1; Table 3), the 25 branch angles were all less than right angles (mean, 39 degrees; range 20 to 64 degrees). None of these junctions resembled the radiation of three or more rhizomes observed from the seedling source.

Examination of the seedling locations indicated that all of the initial rhizomes were still alive and connected at this location. We refer to each of the rhizomes that emanated from this seedling location as the start of a rhizome system, a structure that we use to evaluate some plant characteristics (Fig. 4). Although the largest plant we excavated at the control (Oregon) site reached 4.62 m in total rhizome length, it was small compared to the two plants excavated at the tephra site (Table 3). Control plants were probably fairly young, as they had no dead shoots (Table 3) and formed small distinct patches. However, shoot ages (as indicated by number of leaf plus leaf-scar whorls) reached at least 18 years (Fig. 4a). On each plant, the rhizome systems emanating from the seedling source varied greatly in total rhizome length (Table 3). Across the five control plants, the number of aerial shoots (Fig. 4b) and the maximum distance along rhizomes (Fig. 4c) were related to the total length of a rhizome system.

The two plants we excavated under the tephra had a remarkable extent of interconnected rhizomes, with a total rhizome length of 56.7 and 35.2 m (Table 3). All of the rhizome systems were connected at the source. These two plants almost completely overlapped, forming a single patch over 5 m across; their seedling sources were only 22 cm apart. The rhizomes were of similar color and size throughout and showed no evidence of aging or change in diameter with age. We suspect that these plants and their rhizomes were very old. Although the rhizomes could not be aged and we did not age ramets on tephra plants, we can estimate their age based on

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length of their rhizome systems and ramet ages of control plants. The control plant with the oldest aerial shoot (Control-5; Table 3) was at least 18 years old with a maximum rhizome length of 108 cm from the source. The largest tephra plant (Tephra-1) had a maximum rhizome length about 3 times as great (317 cm; Table 3), which could indicate an age approximately 3 times as great, 50+ years, if rhizome growth rates were similar. Plant Control-4 (Table 3) had a 17-year-old shoot and a maximum rhizome length of 58 cm from the source; based on this growth rate, plant Tephra-1 could be over 90 years old. Estimating exact age is equivocal, but it is clear that rhizomes of *C. umbellata* can survive for many decades. All of the rhizome systems on these large plants at the tephra site were persistent; we saw no evidence of dead rhizome or scars at the source, indicating that rhizomes consistently have long survival. We suspect that the rhizome systems comprising a single plant were all of similar age, given that the number of rhizome systems was similar (3-5; Table 3) instead of increasing on larger plants. However, as with the control plants, the rhizome systems did vary greatly in total rhizome length, ranging from 108-3700 cm for plant Tephra-1 and 150-2556 cm for plant Tephra-2 (Table 3).

The rhizomes were shallow on all plants, from 0-15 cm in depth, and small roots occurred all along the rhizomes. These small roots radiated in all directions and were distinct from the larger, vertical root at the seedling source. Based on the eight rhizome segments measured on plant Tephra-2, small roots averaged 44-66 per meter of rhizome and only 1.2-3.5 cm in length. The root system was shallow; the maximum depth we found was only about 30 cm, even when including the root at the seedling source, and almost all of the root system was within 20 cm of the surface of pre-eruption soil.

Although the rhizome systems are the major component of plant structure, shoots comprise a substantial part of total plant mass (Table 4). The rhizomes are fairly light, generally

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less than 1 g dry mass per m of length, including the roots. The rhizomes are generally >80% of total belowground mass (Table 4), reflecting the small size of roots. Although the two large plants in tephra had a much greater belowground mass than the five control plants, their mass per unit length of rhizome and their leaf mass were similar to those of control plants (Table 4). Stem mass was greater on the large tephra plants than on the control plants, but about 80% of the stem was buried in the tephra. We suspect that tephra plants lost many of their leaves after burial, given the prevalence of buried stem, which likely had leaves when buried. The control plants had consistently less than one-third of their total mass belowground, whereas the two tephra plants had over 80% of their mass belowground (Table 4).

In contrast to *X. tenax*, *C. umbellata* increased substantially in density, frequency, and cover throughout the study period (Fig. 2). Density increased significantly from 1981 to 1983, but in contrast to *X. tenax*, continued to increase from 1983 to 2016 with an over 3-fold increase in both plot types (Table 2). Cover increased 8-fold from 1983-2020 in both plot types; these changes were highly significant (Table 2). However, both cover and density did not differ significantly between natural and cleared plots in either 1981, 1983 or at the end of the study (Table 2). The increase in both density and frequency was abrupt initially, but then continued gradually over the entire period (Fig. 2). The cover increase was delayed relative to density, probably because new ramets took a few years to develop a substantial leaf cover. The continuing density increase in both tephra and cleared plots (Fig. 2a), indicates that this increase is not simply a delayed response to burial. In repeat photos, *C. umbellata* can be seen absent in 1980, but obvious by 2016 when density and cover had increased (Fig. 5).

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Discussion

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We demonstrate that two common understory species have an extreme morphology for forest plants, but in different ways: exceptionally long roots or unusually persistent rhizomes. We show by our examples that they can develop belowground structures beyond the range reported for other forest species. Using a large sample size of permanent plots monitored for 40 years, we demonstrate that these two species differ in population dynamics in ways that follow from their morphological differences. Below we discuss the unusual characteristics of these two species and the implications for generalizations about forest herbs.

Xerophyllum tenax has an unusual aboveground morphology for a forest herb, with dense clumps of narrow strap-like evergreen leaves, unlike those of any other forest herb in the region, a growth form that resembles some desert plants. Although the species grows in a variety of habitats, including open areas, it can be a dominant forest understory plant; in the old-growth forest where we tracked density, it is the most abundant herb (Zobel and Antos 2009). However, the plants are nearly absent in habitats with deeper tephra near Mount St. Helens, probably because the new, elongating, flexible leaves have little potential to push up through tephra and burial is promoted when tephra falls on snow (Antos and Zobel 1982). In contrast to tephra disturbance, *X. tenax* plants survive fire well and sexual reproduction can be promoted by fire (Shebitz et al. 2008, 2009; Peter et al. 2017). Outcrossing appears necessary for seed production (Vance et al. 2004) and plants bloom very little in the shade (Zobel and Antos 2016; Peter et al. 2017), resulting in limited seed production in the understory. In general, seedling establishment is rare to nonexistent in the understory (Peters et al. 2017), which is consistent with our observations over 40 years in an old-growth forest. Although plants may live for at least 60 years (Peter et al. 2017), we propose that they live much longer**,** based on our observation of almost no mortality or sexual recruitment over 40 years, despite a reduction of competition and

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presence of a mineral seedbed provided by tephra. If they establish mostly after fire, as proposed (Peter et al. 2017), that would be over 400 years ago at site SP. Similarly to some other forest herbs, *X. tenax* maintains its populations with almost no sexual reproduction; however, the mechanism differs among species, as *X. tenax* genets persist with little vegetative spread whereas other forest herbs with long-lived genets achieve extensive spread via long rhizomes (e.g., Antos and Zobel 1984).

Xerophyllum tenax can potentially utilize resources from the soil over large areas, as is typical for many forest herbs, but in a way that is unusual (via large root systems rather than by persistent long rhizomes). Many forest herbs have large, persistent rhizome systems (clonal fragments) (Antos and Zobel 1984, 1985*b*; Lezberg et al. 1999, 2001; Nelson et al. 2007), such as *C. umbellata*, which may integrate resources over substantial areas (many m^2), whereas *X. tenax* has rhizomes that, although persistent, are short. Roots are small on most forest herbs (e.g., Antos and Zobel 1984; Lezberg et al. 1999) whereas the extensive system of large roots on *X. tenax* far exceeds in root spread anything found on the many species of forest herbs observed in the region (Antos and Zobel 1984, 1985*b*, Antos 1988; Lezberg et al. 1999, 2001; Nelson et al. 2007; Six and Halpern 2008). The large root system of *X. tenax* also means that the plants are much larger belowground than aboveground; thus it is important not to assume that the size of a plant belowground resembles that of the plant aboveground. In summary, we propose that *X. tenax* has limited vegetative spread via the short rhizomes, but can integrate soil resources over large areas via their extensive root system and can survive for very long periods, probably hundreds of years.

Chimaphila umbellata that we excavated differ from other long-rhizomatous species in the forest understory in that all the rhizome systems retained the structures produced by their

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seedling source. Thus, each plant represented an entire genet and the rhizomes are exceptionally long lived. In contrast, other long-rhizomatous, herbaceous species excavated in the forest understory in the Pacific Northwest almost invariably had clonal fragments that developed from what is now dead rhizome tissue and are only part of a genet (Antos and Zobel 1984; Antos 1988; Lezberg et al. 1999, 2001; Nelson et al. 2007).

Chimaphila umbellata differs from most forest understory species not only in rhizome properties, but also in seedling development. The seeds of this species are very small, dust seeds that depend on fungi for germination and initial development (Eriksson and Kainulainen 2011; Johansson and Eriksson 2013; Lundell et al. 2015). We found no indication of an aerial shoot at the seedling source on any plant we examined. The nearest aerial shoot on the control plants averaged 7 cm (range 5-12 cm) from the vertical root at the seedling source, indicating that early plant development relied on fungal resources and not photosynthate. Thus the plants appear to be initially mycoheterothrophic but develop green leaves subsequently and probably can be considered partially mycoheterothrophic (Lundell et al. 2015), which is the case for almost all Pyroleae (Hynson et al. 2009; Matsuda et al. 2012; Jacquemyn and Merckx 2019). The plants may be fully supported by photosynthate when mature (Hynson et al. 2012; Johansson et al. 2015), but in general green leaves may or may not be beneficial for partially mycoheterothrophic (mixotrophic) plants (Roy et al. 2013).

In any case, the reliance on fungi for seedling establishment may allow initiation from seed in very dark forest understory (Tedersoo et al. 2007) and the extensive and long-lived rhizome systems may allow the plants to spread and persist in adverse places in the understory (Lundell et al. 2015). In addition, the plants might rely on fungi during adverse periods, allowing persistence. *Chimaphyla* is a widespread genus of forest understories that has a long

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history (Liu et al. 2019), which implies that partial mycoheterotropy is a nutritional strategy that has long been an aspect of the diversity of the forest understories. However, we suggest that the exceptionally extensive and persistent rhizome systems of *C. umbellata* represent an equally interesting characteristic, further extending the variation among plants of the forest understory.

Understory herbs differ in many respects including response to disturbance (Roberts 2004), which is nicely illustrated by the large difference between *X. tenax* and *C. umbellata* in their change in abundance after the tephra disturbance (Fig. 2). *Xerophyllum tenax* density and frequency changed only a little early in the study, probably as small buried shoots emerged. This population stability, which we have noted previously (Zobel and Antos 2009), is consistent with the species' height and accompanying high survival, long-lived ramets, and limited vegetative spread. In contrast, *C. umbellata* increased greatly in density, frequency and cover, which is likely the result of the growth of new long rhizomes and aerial shoots that allowed the plants not only to increase in density and cover but also to enter new plots and thus increase in frequency. Of the plots that *X. tenax* occupied 25 years after the tephra fall, it already occupied 96% of these plots immediately after the tephra fall, whereas the corresponding value for *C. umbellata* was only 54% (Zobel and Antos 2009). The flexibility in growth related to rhizomatous spread allowed *C. umbellata* to take advantage of the altered conditions after disturbance. Curiously *C. umbellata* increased not only in tephra plots at Mount St. Helens, but also in experimentally cleared plots. The large interconnected rhizome systems often must extend outside the 1 m^2 cleared plots, making the response in cleared plots partially dependent on parts of the plant still affected by tephra; altered soil conditions (Zobel and Antos 1991) or reduced competition may have produced a general increase in *C. umbellata* growth. Damage to leaves and stems by tephra could explain the much higher proportion of biomass belowground on the large tephra plants

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compared to the smaller control plants, but an ontogenetic shift is an alternative explanation. The large number of growing rhizome tips on plants in the tephra (Table 3) could be a response to damage. However, an ontogenetic change with plant size is supported by the number of shoots per meter of rhizome (2.7-6.1 for the small control plants but only 0.23 and 0.46 for the two much larger plants in the tephra). These values include both live and dead shoots, so damage from tephra cannot be the only factor responsible for this large difference. In general, the morphological response of plants to tephra burial varied greatly (Antos and Zobel 1985*a*, *b*) as did their subsequent change in abundance during the years after the disturbance (Zobel and Antos 1997, 2009, 2017; Fischer et al. 2019). Even in very old forests the understory is dynamic and changes can occur over long periods even in response to minor disturbances (e.g., for tephra, Fischer et al. 2016; Zobel and Antos 2017).

Forest understory herbs are far from a homogeneous group. They are highly diverse, with species that differ in a variety of characteristics, including belowground structure. Genets spread little or not at all in some (e.g., *Trillium ovatum*, Jules 1998) whereas most species are clonal (Bierzychudek 1982; Whigham 2004). Many species have extensive clonal spread via rhizomes or stolons and the current clonal fragments typically originate in what is now dead rhizome or stolon, implying that the genet may be much older than the live tissue (Antos and Zobel 1984; Lezberg et al. 1999). However, rhizome connections can be very short lived (e.g., *Trientalis europaea*, Piqueras and Klimeš 1998) or persist for many years**,** resulting in large interconnected systems (e.g., *Achlys triphylla*, *Clintonia uniflora*, Antos and Zobel 1984). The forest understory environment is highly heterogeneous (Burton et al. 2011; Chudomelová and Zelený 2020), and large interconnected rhizome and stolon systems should allow integration of resources and persistence of individual shoots in adverse microsites. Although large clonal

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fragments occur in many species, *C. umbellata* appears to have rhizomes that are more persistent and connected than in any other species examined, and is unique among long rhizomatous species in the persistence of a seedling source (consistently present in the plants we excavated versus rare in other species). *Xerophyllum tenax* is also unusual among forest understory plants in the extent of its root system. Thus it can potentially integrate supplies of water and nutrients (but not photosynthate) over large areas, although not via the long and interconnected rhizome or stolon systems typical of many species in Pacific Northwest forests (Antos and Zobel 1984; Lezberg et al. 1999, 2001; Nelson et al. 2007). Although *X. tenax* and *C. umbellata* differ greatly in belowground structure, they are similar in some respects. They both retained connections between ramets and flowered infrequently compared to most species within a group of 48 understory species in our study sites (Zobel and Antos 2016); among these herbs, those with more stable populations generally flowered less frequently (Zobel and Antos 2009). They are both evergreen with leaves that persist for a number of years in contrast to many forest herbs with deciduous leaves, which may make resource integration over long periods especially important. These two species have exceptionally large belowground structures that could facilitate integration of soil resources, but these structures are very different.

Forest understory herbs differ greatly in many life history characteristics, for example, traits that can affect responses to disturbance (e.g., Halpern 1989; Roberts 2004; Willson et al. 2018) or amounts of seedling establishment, which can be especially important in some situations (Matlack and Monde 2004; Albrecht and McCarthy 2009; Gaudeul et al. 2019). Both of the two species we studied here lacked effective seedling establishment following the tephra disturbance, but they differed in their post-disturbance changes in importance. *Xerophyllum tenax* showed only minor increases, which is explicable in terms of the limited ability to spread

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via the short rhizomes, whereas *C. umbellata* increased greatly in both cover and density, which is likely related to the extensive rhizome system that could produce many new aerial shoots. We suggest that the differences in post-disturbance population dynamics are a function of the belowground structure of these two species and that the most important variation among forest herbs occurs in belowground structures; species with similar aboveground morphology can differ greatly belowground (Antos 1988). Here we extend the range of variation we have previously reported in the belowground structure of forest understory herbs. We propose that the impressive range of variation among forest herbs allows species to exploit the wide range of environmental variation across space and time, both of light and soil resources occurring in the forest understory (e.g., Burton et al. 2011; Chudomelová and Zelený 2020), including alterations due to disturbance (e.g., Peterson et al. 1990; Kern et al 2019). We also suggest that species can exploit similar microenvironments in different ways (e.g., resource integration via large root systems of *X. tenax* versus long rhizomes and large clonal fragments of other species). To understand the dynamics of forest understory plants, and their critical role in forest ecosystems, it is essential to appreciate the extensive variation among these species, especially in their belowground morphology.

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References

Albrecht, M.A., and McCarthy, B.C. 2009. Seedling establishment shapes the distribution of shade-adapted forest herbs across a topographical moisture gradient. J. Ecol. **97**(5): 1037-1049. doi: 10.1111/j.1365-2745.2009.01527.x.

Antos, J.A. 1988. Underground morphology and habitat relationships of three pairs of forest herbs. Am. J. Bot. **75**(1): 106-113. doi:10.2307/2443908.

Antos, J.A., and Zobel, D.B. 1982. Snowpack modification of volcanic tephra effects on forest understory plants near Mount St. Helens. Ecology **63**(6): 1969-1972. doi:10.2307/1940135.

Antos, J.A., and Zobel, D.B. 1984. Ecological implications of belowground morphology of nine coniferous forest herbs. Bot. Gaz. **145**(4): 508-517.

Antos, J.A., and Zobel, D.B. 1985*a*. Plant form, developmental plasticity, and survival following burial by volcanic tephra. Can. J. Bot. **63**(12): 2083-2090. doi:10.1139/b85-293.

Antos, J.A., and Zobel, D.B. 1985*b*. Upward movement of underground plant parts into tephra deposits from Mount St. Helens. Can. J. Bot. **63**(12): 2091-2096. doi:10.1139/b85-294.

Antos, J.A., and Zobel, D.B. 1985*c*. Recovery of forest understories buried by tephra from Mount St. Helens. Vegetatio **64**(2-3): 103-111. doi:10.1007/BF00044786.

Antos, J.A., and Zobel, D.B. 1986. Seedling establishment in forests affected by tephra from Mount St. Helens. Am. J. Bot. **73**(4): 495-499. doi:10.2307/2444253.

Barbier, S., Gosselin, F., and Balandier, P. 2008. Influence of tree species on understory vegetation diversity and mechanisms involved – a critical review for temperate and boreal forests. For. Ecol. Manage. **254**(1): 1-15. doi:10.1016/j.foreco.2007.09.038.

Draft

© The Author(s) or their Institution(s)

Bierzychudek, P. 1982. Life histories and demography of shade-tolerant temperate forest herbs: a review. New Phytol. **90**(4):757-776.

Burton, J.I., Mladenoff, D.J., Clayton, M.K., and Forrester, J.A. 2011. The roles of environmental filtering and colonization in the fine-scale spatial patterning of ground-layer plant communities in north temperate deciduous forests. J. Ecol. **99**(3):764-776. doi: 10.1111/j.1365-2745.2011.01807.x.

Cain, M.L., Damman, H., and Muir, A. 1998. Seed dispersal and the Holocene migration of woodland herbs. Ecol. Monogr. **68**(3): 325-347. doi:10.1890/0012.9615(1998)068[0325:SDATHM]2.0.CO;2.

Chamagne, J., Paine, C.E.T., Schoolmaster, Jr., D.R., Stejskal, R., Volařik, D., Ŝebesta, J., et al. 2016. Do the rich get richer: Varying effects of tree species identity and diversity on the richness of understory taxa. Ecology **97**(9): 2364-2373. doi:10.1002/ecy.1479.

Chudomelová, M., and Zelený, D. 2020. Tracing the signs of local dispersal in the temperate forest understorey using spatially structured vegetation data. J. Veg. Sci. **31**(1): 84-94. doi:10:1111/jvs.12835.

Coomes, D.A., and Grubb, P.J. 2000. Impacts of root competition in forests and woodlands: A theoretical framework and review of experiments. Ecol. Monogr. **70**(2): 171-207. doi:10.1890/0012.9615(2000)070[0171:IORCIF]2.0.CO;2.

Ehrlén, J., and Eriksson, O. 2000. Dispersal limitation and patch occupancy in forest herbs. Ecology **81**(6): 1667-1674. doi:10.1890/0012.9658(2000)081.

Eriksson, O., and Kainulainen, K. 2011. The evolutionary ecology of dust seeds. Perspect. Plant Ecol. Evol. Syst. **13**(2): 73-87. doi:10.1016/j.ppees.2011.02.002.

Draft

© The Author(s) or their Institution(s)

Fischer, D.G., Antos, J.A., Grandy, W.G., and Zobel, D.B. 2016. A little disturbance goes a long way: 33-year understory successional responses to a thin tephra deposit. For. Ecol. Manage. **382**: 236-243. doi:10.1016/j.foreco.2016.10.018.

Fischer, D.G., Antos, J.A., Zobel, D.B., and Biswas, A. 2019. Understory succession after burial by tephra from Mount St. Helens. J. Ecol. **107**(2): 531-544. doi.org/10.1111/1365-2745.13052.

Franklin, J.F., and Dyrness, C.T. 1973. Natural vegetation of Oregon and Washington. U.S. Dep. Agri. For. Serv. Gen. Tech. Rep. PNW-8.

Gaudeul, M., Delahaye, T., and Muller, S. 2019. AFLP markers show low levels of clonal propagation and high genotypic diversity in the rare, southernmost populations of *Linnaea borealis* L. (Caprifoliaceae) in the Western Alps. Genetica **147**(1):79-90. doi.org/10.1007/s10709-019-00054-6.

Gilliam, F.S. 2007. The ecological significance of the herbaceous layer in temperate forest ecosystems. BioScience **57**(10): 845-858. doi:10.1641/B571007.

Halpern, C.B. 1989. Early successional patterns of forest species: Interactions of life history traits and disturbance. Ecology **70**(3): 704-720. doi:10.2307/1940221.

Hitchcock, C.L., and Cronquist, A. 2018. Flora of the Pacific Northwest. Second edition edited by D.E. Giblin, B.S. Legler, P.F. Zika, and R.G. Olmstead. University of Washington Press, Seattle, Washington.

Hynson, N.A., Preiss, K., Gebauer, G., and Bruns, T.D. 2009. Isotopic evidence of full and partial myco-heterotrophy in the plant tribe Pyroleae. New Phytol. **182**(3):719-726. doi:10.1111/j.1469-8137.2009.027.

Draft

© The Author(s) or their Institution(s)

Hynson, N.A., Mambelli, S., Amend, and Dawson, T.E. 2012. Measuring carbon gains from fungal networks in understory plant from the tribe Pyroleae (Ericaceae): a field manipulation and stable isotope approach. Oecologia **169**(2):307-317. doi:10.1007/s00442-011-2198-3.

Jacquemyn, H., and Merckx, V.S.F.T. 2019. Mycorrhizal symbioses and the evolution of trophic modes in plants. J. Ecol. **107**(4):1567-1581. doi:10.1111/1365-2745.13165.

Johansson, V.A., and Eriksson, O. 2013. Recruitment limitation, germination of dust seeds, and early development of underground seedlings in Pyroleae species. Botany **91**(1): 17-24. doi.org/10.1139/cjb-2012-0153.

Johansson, V.A., Mikusinska, A., Ekbald, A., and Eriksson, O. 2015. Partial mycoheterotrophy in Pyroleae: nitrogen and carbon stable isotope signatures during development from seedling to adult. Oecologia **177**(1): 203-211. doi:10.1007/s00442-014-3137-x.

Jules, E.S. 1998. Habitat fragmentation and demographic change for a common plant: *Trillium* in old-growth forest. Ecology **79**(5): 1645-1656. doi:10.1890/0012-9658(1998)079.

Kern, C. C., Schwarzmann, J., Kabrick, J., Gerndt, K., Boyden, S., and Stanovick, J.S. 2019. Mounds facilitate regeneration of light-sensitive and browse-sensitive tree species after moderate severity wind disturbance. For. Ecol. Manage. **437**:139-147. doi:10.1016/j.foreco.2018.12.040.

Lezberg, A.L., Antos, J.A., and Halpern, C.B. 1999. Below-ground traits of herbaceous species in young, coniferous forests of the Olympic Peninsula, Washington. Can. J. Bot. **77**(7):936-943. doi:10-1139/b99-056.

Draft

© The Author(s) or their Institution(s)

Lezberg, A.L., Halpern, C.B., and Antos, J.A. 2001. Clonal development of *Maianthemum dilatatum* in forests of differing age and structure. Can. J. Bot. **79**(9):1028-1038. doi:10-1139/b01-083.

Lindh, B.C. 2005. Effects of conifer basal area on understory herb presence, abundance, and flowering in a second-growth Douglas-fir forest. Can. J. For. Res. **35**(4): 938-948. doi:10.1139/X05-027.

Lindh, B.C., Gray, A.N., and Spies, T.A. 2003. Responses of herbs and shrubs to reduced root competition under canopies and in gaps: a trenching experiment in old-growth Douglas-fir forests. Can. J. For. Res. **33**(10): 2052-2057. doi:10.1139/X03-120.

Liu, Z.-W., Zhou, J., Peng, H., Freudenstein, J.V., and Milne, H.I. 2019. Relationships between Tertiary relict and circumboreal woodland floras: a case study in *Chimaphila* (Ericaceae). Ann. Bot. **123**(6): 1089-1098. doi:10.1093/aob/mcz018.

Lundell, A., Cousins, S.A.O., Eriksson, O. 2015. Population size and reproduction in the declining endangered forest plant *Chimaphila umbellata* in Sweden. Folia Geobot. **50**(1):13-23. doi:10.1007/s12224-015-9212-1.

Matlack, G.R. 2005. Slow plants in a fast forest: local dispersal as a predictor of species frequencies in a dynamic landscape. J. Ecol. **93**(1): 50-59. doi:10.1111/j.1365-2745.2004.00947.x.

Matlack, G.R., and Monde, J. 2004. Consequences of low mobility in spatially and temporally heterogeneous ecosystems. J. Ecol. **92**(6): 1025-1035. doi:10.1111/j.0022-0477.2004.00908.x.

Matsuda, Y., Shimizu, S., Mori, M., Ito, S.-I., and Selosse, M.-A. 2012. Seasonal and environmental changes of mycorrhizal associations and heterothrophy levels in

Draft

© The Author(s) or their Institution(s)

mixotrophic *Pyrola japonica* (Ericaceae) growing under different light environments. Am. J. Bot. **99**(7): 1177-1188. doi:10.3732/ajb.1100546.

McCune, B. 1986. Root competition in a low-elevation grand fir forest in Montana: a trenching experiment. Northwest Sci. **60**(1):52–54.

Moola, F.M., and Vasseur, L. 2009. The importance of clonal growth to the recovery of *Gaultheria procumbens* L. (Ericaceae) after forest disturbance. Plant Ecol. **201**(1): 319-337. doi:10.1007/s11258-008-9496-9.

Nelson, C.R., Halpern, C.B., and Antos, J.A. 2007. Variation in responses of late-seral herbs to disturbance and environmental stress. Ecology **88**(11): 2880-2890. doi:10.1890/06-1989.1.

Peter, D.H., Harrington, T.B., and Thompson, M. 2017. Effects of the light environment and stand history on beargrass (*Xerophyllum tenax*) morphology and demography. Northwest Sci. **91**(4):367-382. doi:10.3955/046.091.0406.

Peterson, C. J., Carson, W.P., McCarthy, B.C., and Pickett, S.T.A. 1990. Microsite variation and soil dynamics within newly created treefall pit and mounds. Oikos **58**(1): 39-46. doi:10.2307/3565358.

Piqueras, J., and Klimeš, L. 1998. Demography and modelling of clonal fragments in the pseudoannual plant *Trientalis europaea* L. Plant Ecol. **136**(2): 213-227. doi: 10.1023/A:1009753515163.

Roberts, M.R. 2004. Response of the herbaceous layer to natural disturbance in the herbaceous layer of North American forests. Can. J. Bot. **82**(9): 1273–1283. doi:10.1139/B04-091.

Draft

© The Author(s) or their Institution(s)

Roy, M., Gonneau, C., Rocheteau, A., Berveiller, D., Thomas, J.-C., Damesin, C., and Selosse, M.-A. 2013. Why do mixotrophic plants stay green? A comparison between green and acholoropyllous orchids. Ecol. Monogr. **83**(1): 97-117. doi:10.1890/11-2120.1.

Shebitz, D.J., Reichard, S.H., and Woubneh, W. 2008. Beargrass (*Xerophyllum tenax*) on the Olympic Peninsula, Washington: autecology and populations status. Northwest Sci. **82**(2): 128-140. doi:10.3955/0029-344X-82.2.128.

Shebitz, D.J., Reichard, S.H., and Dunwiddie, P.W. 2009. Ecological and cultural significance of burning beargrass habitat on the Olympic Peninsula, Washington. Ecol. Restor. **27**(3): 306-319. doi:10.3368/er.27.3.306.

Six, L.J., and Halpern, C.B. 2008. Substrate effects on distribution, biomass allocation, and morphology of forest understory plants. Botany **86**(10):1133-1142. doi:10.1139/B08-086.

Slade, A.J., and Hutchings, M.J. 1987. Clonal integration and plasticity in foraging behavior in *Glechoma hederacea*. J. Ecol. **75**(4):1023-1036.

Tedersoo, L., Pellet, P., Koljalg, U., and Selosse, M.A. 2007. Parallel evolutionary paths to mycoheterotrophy in understorey Ericaceae and Orchidaceae: ecological evidence for mixotrophy in Pyroleae. Oecologia **151**(2): 206-217. doi:10.1007/s00442-006-0581-2.

Valverde, T., and Silvertown, J. 1998. Variation in the demography of a woodland herb (*Primula vulgaris*) along the forest regeneration cycle: projection matrix analysis. J. Ecol. **86**(4): 545-562. doi:10.1046/j.1365-2745.1998.00280.x.

van Groenendael, J.M., Klimes, L., Klimesova, J., and Hendriks, R.J.J. 1996. Comparative ecology of clonal plants. Philos. Trans. R. Soc. Lond. B **351**(1345): 1331-1339.

Draft

© The Author(s) or their Institution(s)

Vance, N.C., Bernhardt, P., and Edens, R.M. 2004. Pollination and seed production in *Xerophyllum tenax* (Melanthiaceae) in the Cascade Range of Central Oregon. Amer. J. Bot. **91**(12): 2060-2068. doi:10.3732/ajb.91.12.2060.

Verheyen, K., Guntenspergen, G.R., Biesbrouck, B., and Hermy, M. 2003. An integrated analysis of the effects of past land use on forest herb colonization at the landscape scale. J. Ecol. **91**(5): 731-742. doi:10.1046/j.1365-2745.2003.00807.x.

Verheyen, K., and Hermy, M. 2004. Recruitment and growth of herb-layer species with different colonizing capacities in ancient and recent forests. J. Veg. Sci. **15**(1): 125-134. Doi:10:1111/j.1654-1103.2004.tb02245.x.

Whigham, D.F. 2004. Ecology of woodland herbs in temperate deciduous forests. Annu. Rev. Ecol. Evol. Syst. **35**: 583-621. doi:10.1146/annurev.ecolsys.35.021103.105708.

Willson, K.G., Barefoot, C.R., Hart, J.L., Schweitzer, C.J., and Dey, D.C. 2018. Temporal patterns of ground flora response to fire in thinned *Pinus-Quercus* stands. Can. J. For. Res. **48**(10): 1171-1183. doi.org/10.1139/cjfr-2018-0132.

Zobel, D.B., and Antos, J.A. 1987. Composition of rhizomes of forest herbaceous plants in relation to morphology, ecology and burial by tephra. Bot. Gaz. **148**(4): 490-500.

Zobel, D.B., and Antos, J.A. 1991. 1980 tephra from Mount St. Helens: Spatial and temporal variation beneath forest canopies. Biol. Fertil. Soils **12**(1): 60-66. doi:10.1007/BF00369389.

Zobel, D.B., and Antos, J.A. 1997. A decade of recovery of understory vegetation buried by volcanic tephra from Mount St. Helens. Ecol. Monogr. **67**(3): 317-344. doi:10.1890/0012-9615(1997)067[0317:ADOROU]2.0.CO;2.

Draft

© The Author(s) or their Institution(s)

Zobel, D.B., and Antos, J.A. 2007. Flowering and seedling production of understory herbs in old-growth forests affected by 1980 tephra from Mount St. Helens. Can. J. Bot. **85**(7): 607-620. doi:10.1139/B07-057.

Zobel, D.B., and Antos, J.A. 2009. Species properties and recovery from disturbance: forest herbs buried by volcanic tephra. J. Veg. Sci. **20**(4): 650-662. doi-org/10.1111/j.1654-1103.2009.01057.x.

Zobel, D.B., and Antos, J.A. 2016. Flowering patterns of understory herbs 30 years after disturbance of subalpine old-growth forests by tephra from Mount St. Helens. Int. J. Plant Sci. **177**(2): 145-156. doi:10.1086/684181.

Zobel, D.B., and Antos, J.A. 2017. Community reorganization in forest understories buried by volcanic tephra. [online] Ecosphere **8**(12):e02045. doi: 10.1002/ecs2.2045

Zobel, D.B., and Antos, J.A. 2018. Forest understory buried by volcanic tephra: inertia, resilience and the pattern of community re-development. *In* Ecological responses at Mount St. Helens: Revisited 35 years after the 1980 eruption. *Edited by* C.M. Crisafulli and V.H. Dale. Springer, New York. pp. 113-125.

Draft

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Table 1. Characteristics of 11 *Xerophyllum tenax* examined at a site with ca. 10 cm tephra about 23 km northeast of Mount St. Helens. ND = no data.

Plant Number	Plant height[a] (cm)	Plant diameter[b] (cm)	Number of crowns[c]	Minimum depth of rhizome in old soil (cm)
1	30	85	12	5
2	21	72	2	ND
3	21	50	4	3
4	30	70	7	5
5	18	42	1	6
6	38	90	35	5
7	25	50	1	7
8	12	22	1	4
9	54	70	13	4
10	37	45	6	6
11	10	30	1	5

[a] Leaf height above the tephra

[b] Diameter of the foliage of a plant. Each crown (or ramet) had leaves but ramets were close together at the ends of the short rhizomes such that the leaves of all crowns together formed a single dense canopy

[c] Rhizome ends with tufts of foliage

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Table 2. Changes in cover (%) and density (number of aerial shoots/m^2) of *Xerophyllum tenax* and *Chimaphila umbellata* from 1981 to 1983 and then from 1983 to 2020 (or 2016) in an old-growth forest that received 4.5 cm of tephra from Mount St. Helens in 1980 (site SP in Fischer et al. 2019). Natural plots had undisturbed tephra (n = 101) and cleared plots (n = 50) had tephra removed during summer 1980. Temporal comparison are based on a paired t-test; significance cut-offs can be considered 0.05/2 because of the two transitions that are tested. Comparisons between natural and cleared plots, based on a t-test for unequal variances, were all non-significant except for *X. tenax* cover in 1981 (P = 0.024). P-values are in bold.

Species and cover or density	Plot type	1981	1983	2020 (except 2016 for *C. umbellata* density)	1981 versus 1983 P-value	1983 versus 2020 (2016) P-value
X. tenax cover	Natural	12.27	16.10	23.14	**0.000**	**0.000**
	Cleared	20.44	24.26	28.54	**0.003**	**0.013**
X. tenax density	Natural	2.58	3.14	3.35	**0.001**	**0.334**
	Cleared	3.06	3.98	3.84	**0.000**	**0.492**
C. umbellata cover	Natural	0.45	0.48	4.14	**0.466**	**0.000**
	Cleared	0.28	0.48	3.85	**0.013**	**0.000**
C. umbellata density	Natural	1.01	1.89	6.89	**0.000**	**0.000**
	Cleared	0.90	1.90	7.04	**0.005**	**0.001**

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Table 3. Characteristics of seven *Chimaphila umbellata* plants (five in a control site and two at a site with ca. 10 cm tephra about 23 km northeast of Mount St. Helens) that were completely excavated.

Site and plant number	Total rhizome length (cm)	Maximum distance from source [a] (cm)	Number of rhizomes from source [b]	Range in rhizome system length[c] (cm)	Number of green aerial shoots	Number of dead aerial shoots	Number of growing rhizome tips
Control-1	147	75	3	7-81	4	0	3
Control-2	462	84	4	32-190	22	0	6
Control-3	214	52	5	11-76	13	0	8
Control-4	211	58	4	10-128	10	0	1
Control-5	211	108	3	6-136	6	0	1
Tephra-1	5674	317	5	108-3700	9	17	42
Tephra-2	3516	283	3	170-2556	4	4	20

[a] Maximum distance along a rhizome from the seedling source location to the farthest rhizome end

[b] Number of rhizomes coming from the seedling source location

[c] Range in total rhizome length for rhizome systems emanating from the seedling source

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Table 4. Dry mass of parts of seven *Chimaphila umbellata* plants (five in a control site and two at a tephra site) that were completely excavated. Plants were dried to constant mass at 70 C. ND = no data.

Site and plant number	Total below-ground mass [a] (g)	Rhizome as proportion of total below-ground mass[b]	*Below-ground* mass per meter of rhizome [c] (g/m)	Stem mass [d] (g)	Leaf mass (g)	Proportion of total mass below-ground [e]
Control-1	0.955	ND	0.650	0.871	1.389	0.297
Control-2	3.626	0.902	0.785	3.867	6.921	0.252
Control-3	1.348	0.801	0.630	1.120	3.183	0.239
Control-4	1.438	ND	0.682	1.532	2.613	0.258
Control-5	1.614	ND	0.765	1.110	2.280	0.323
Tephra-1	59.480	0.893	1.048	10.155	2.165	0.828
Tephra-2	28.280	0.905	0.813	4.010	1.620	0.812

[a] Rhizome plus roots

[b] Total includes roots and rhizomes

[c] Total of root and rhizome mass divided by total rhizome length

[d] Stems are upturned rhizome ends that are aboveground, but for tephra plants much of the stem was buried by tephra (79.3 % of stem mass for Tephra-1 and 81.1 % for Tephra-2)

[e] Stems in tephra are included in aboveground parts; thus belowground includes only rhizomes and roots in the soil (below tephra)

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Figure captions:

Fig. 1. Representative plants of the study species in 2013 at site SP, 58 km northeast of Mount St. Helens, with 4.5 cm tephra: (a) *Xerophyllum tenas.* (b) *Chimaphyla umbellata.*

Fig. 2. Density (a), frequency (b), and cover (c) of *Xerophyllum tenax* and *Chimaphyla umbellata* shoots over 36 or 40 years (1980-2016 or 2020) in an old-growth forest that received 4.5 cm of tephra from Mount St. Helens in 1980 (site SP in Fischer et al. 2019). Values are for 101 natural plots with unaltered tephra (Nat) and 50 plots cleared of tephra in 1980 (Cl). Plots are 1 m^2 and permanently marked for repeat sampling. Density is mean number of crowns for *X. tenax* and number of aerial shoots for *C. umbellata* per m^2 plot and frequency is the percentage of 1 m^2 plots containing any cover for the species. Cover is the visual estimate of percent crown cover. Bars represent 1 SE of the mean. See Table 2 for significance of temporal changes and natural versus cleared plot differences. Xete = *Xerophyllum tenax* and Chum = *Chimaphyla umbellata.*

Fig. 3. Repeat photos from the SP site at Mount St. Helens in 1980 (left) and again in 2016 (right) demonstrating stability in *Xerophyllum tenax* density and ramet location through time. Each arrow indicates one ramet (crown). DWD = downed woody debris.

Fig. 4. Characteristics of five *Chimaphyla umbellata* completely excavated at the control site in the Cascade Mountains of Oregon. (a) Aerial shoot age (as indicated by number of whorls of leaves plus leaf-scars) versus distance along the rhizomes from the seedling source. (b) Number

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of aerial shoots versus total rhizome length for individual rhizome systems emanating from the seedling source. See Table 2 for number of rhizome systems on each plant that come from the seedling source. (c) Maximum distance along a rhizome system versus total length of the rhizome system (including all branch rhizomes).

Fig. 5. Repeat photos of *Chimaphyla umbellata* (indicated by circles) in two plots in 1980 (left), when *C. umbellata* was absent, and again in 2016 (right) when presence of the species is visually obvious.

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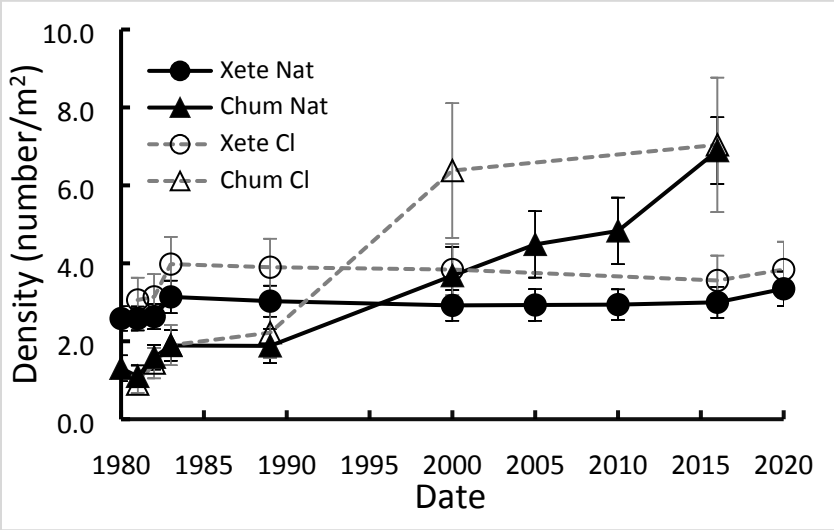
a



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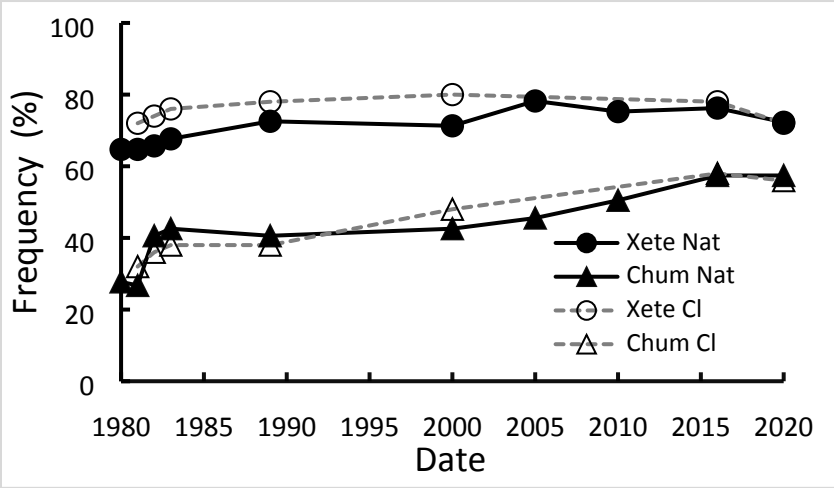

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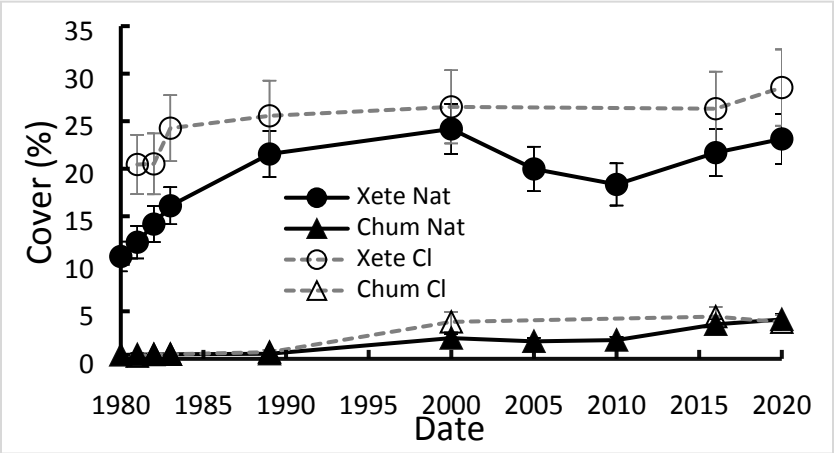

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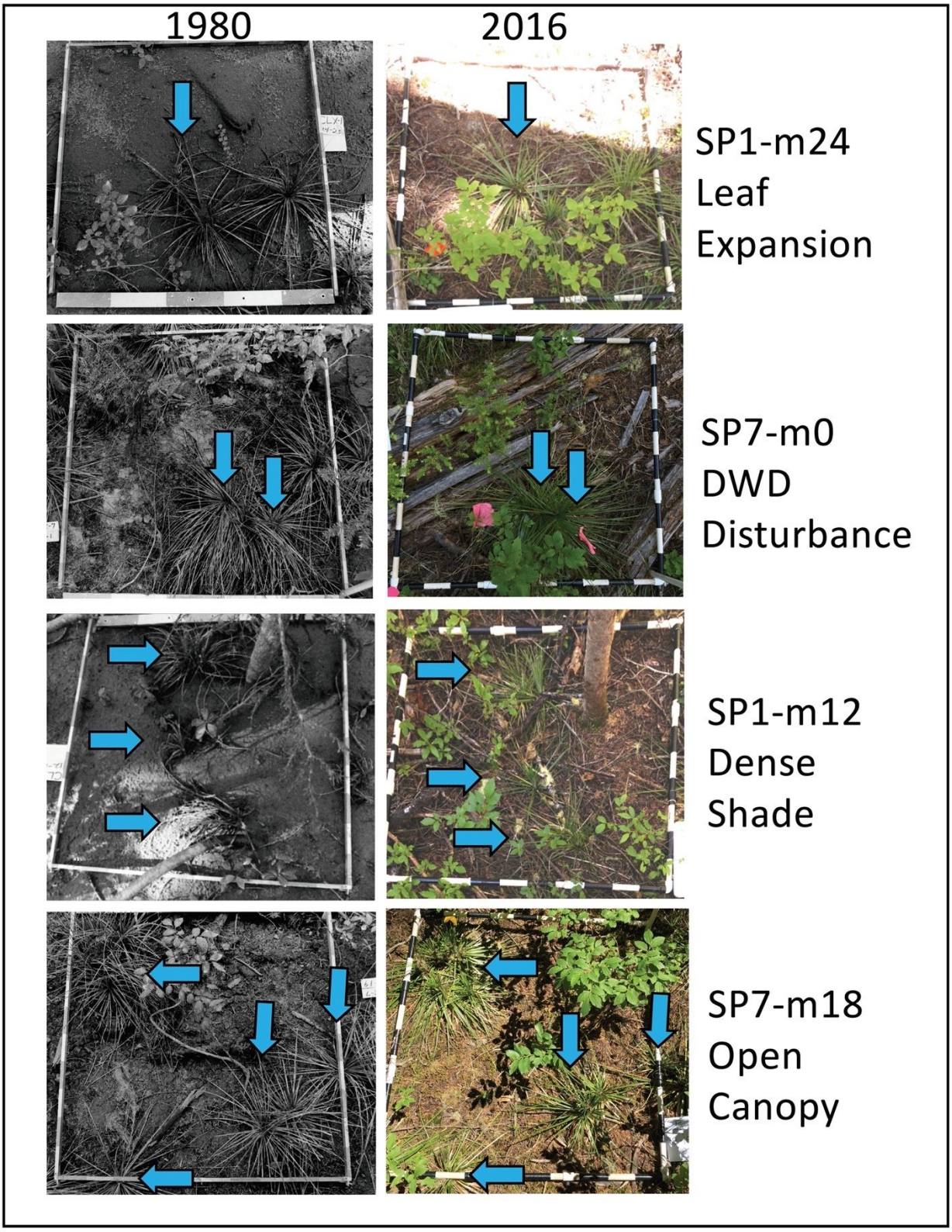

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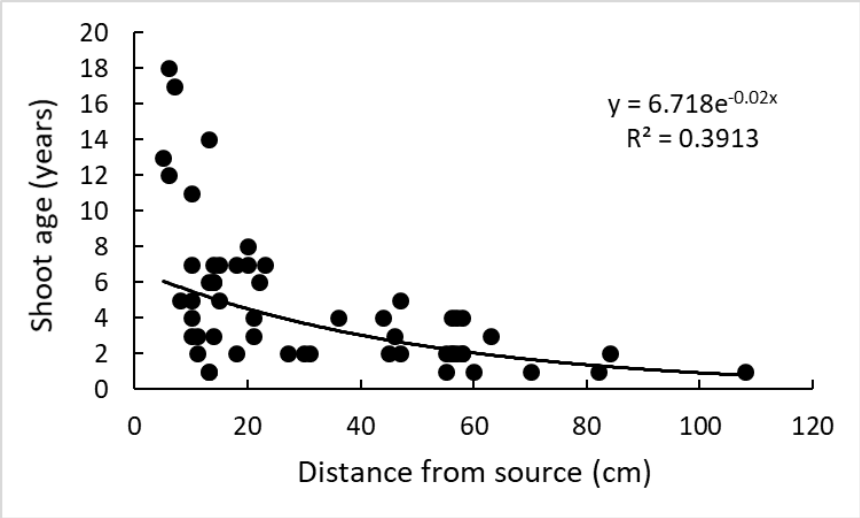
a



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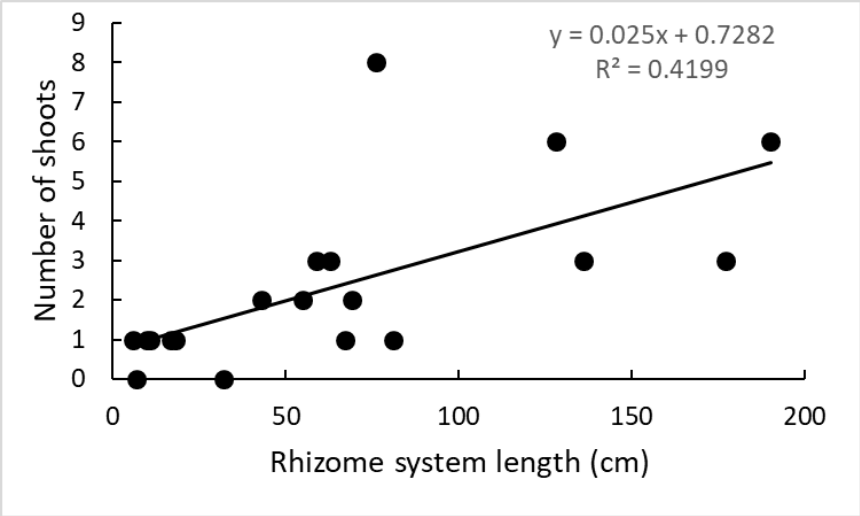

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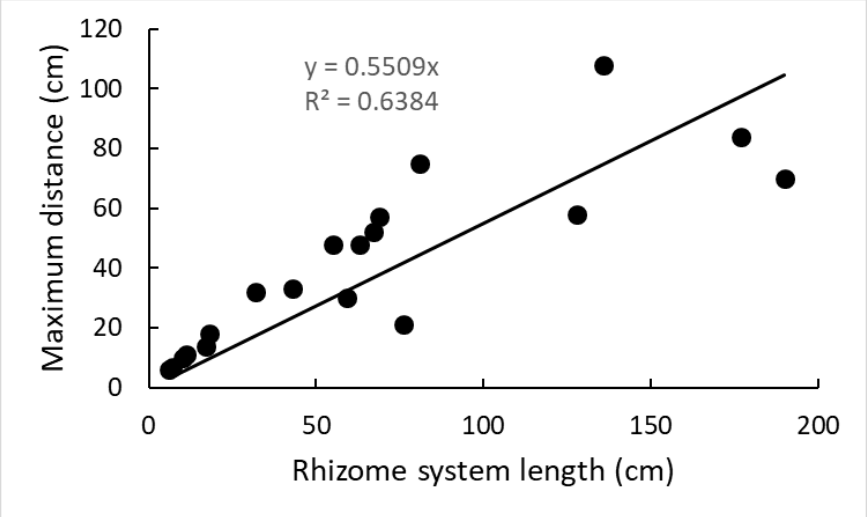

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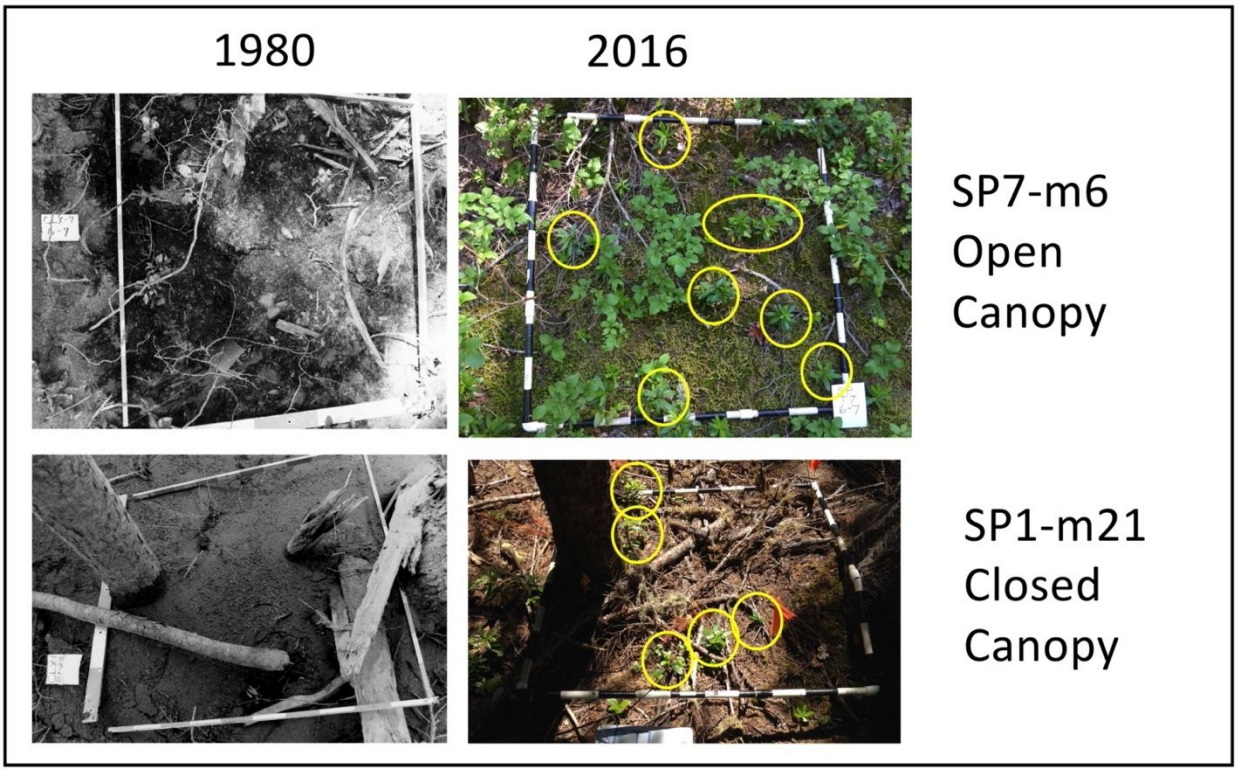

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